Exploratory Data Analysis on Retail Sales Data

Internship Project – Oasis Infobyte

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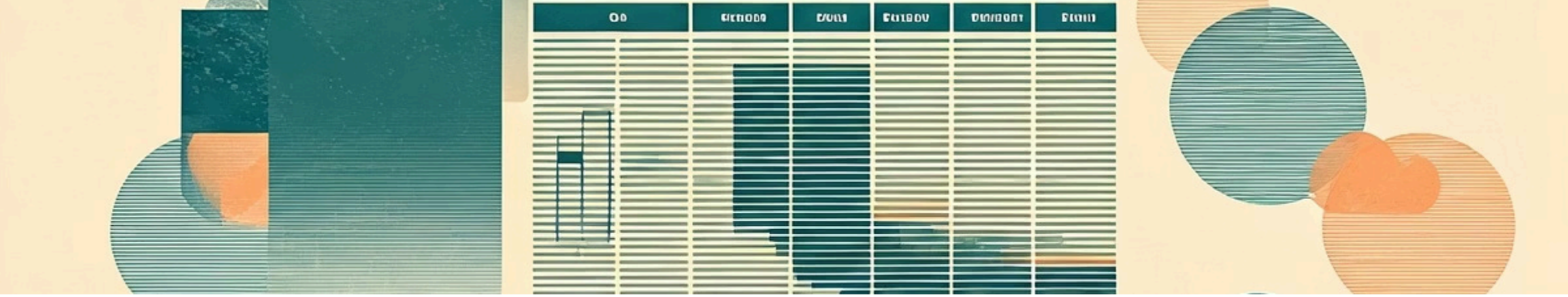

Problem Statement & Dataset

Problem Statement

- Analyze retail sales data
- Identify sales trends and patterns
- Understand customer purchasing behavior
- Provide data-driven recommendations

Dataset Description

- Retail transaction records
- Columns: Date, Sales, Product Category, Customer, Quantity, Revenue
- Includes time-based sales information

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Data Preparation & Statistics

Data Loading & Cleaning

- Loaded dataset using Pandas
- Checked missing values
- Removed duplicates
- Converted date to datetime format
- Verified data types

Descriptive Statistics

- Calculated mean, median, standard deviation
- Analyzed sales distribution
- Identified overall revenue performance

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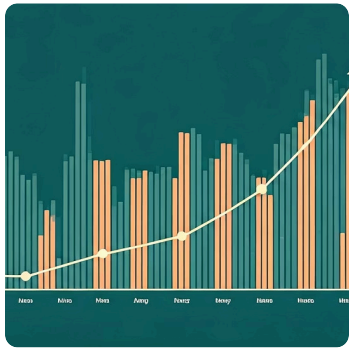
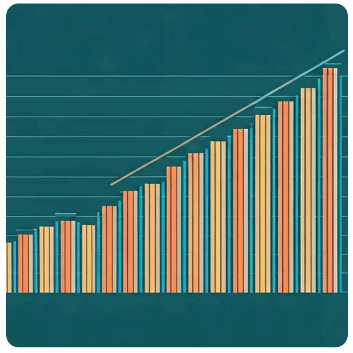
Time Series Analysis



Monthly Sales Trend

Analyzed sales fluctuations over time.



Peak Periods

Identified highest and lowest sales periods.

Seasonal Patterns

Observed recurring seasonal buying behaviors.

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Product & Customer Analysis

Product Analysis

- Identified top-selling product categories
- Compared revenue across categories
- Analyzed quantity vs. revenue

Customer Analysis

- Analyzed purchasing behavior
- Identified repeat customers
- Observed buying patterns

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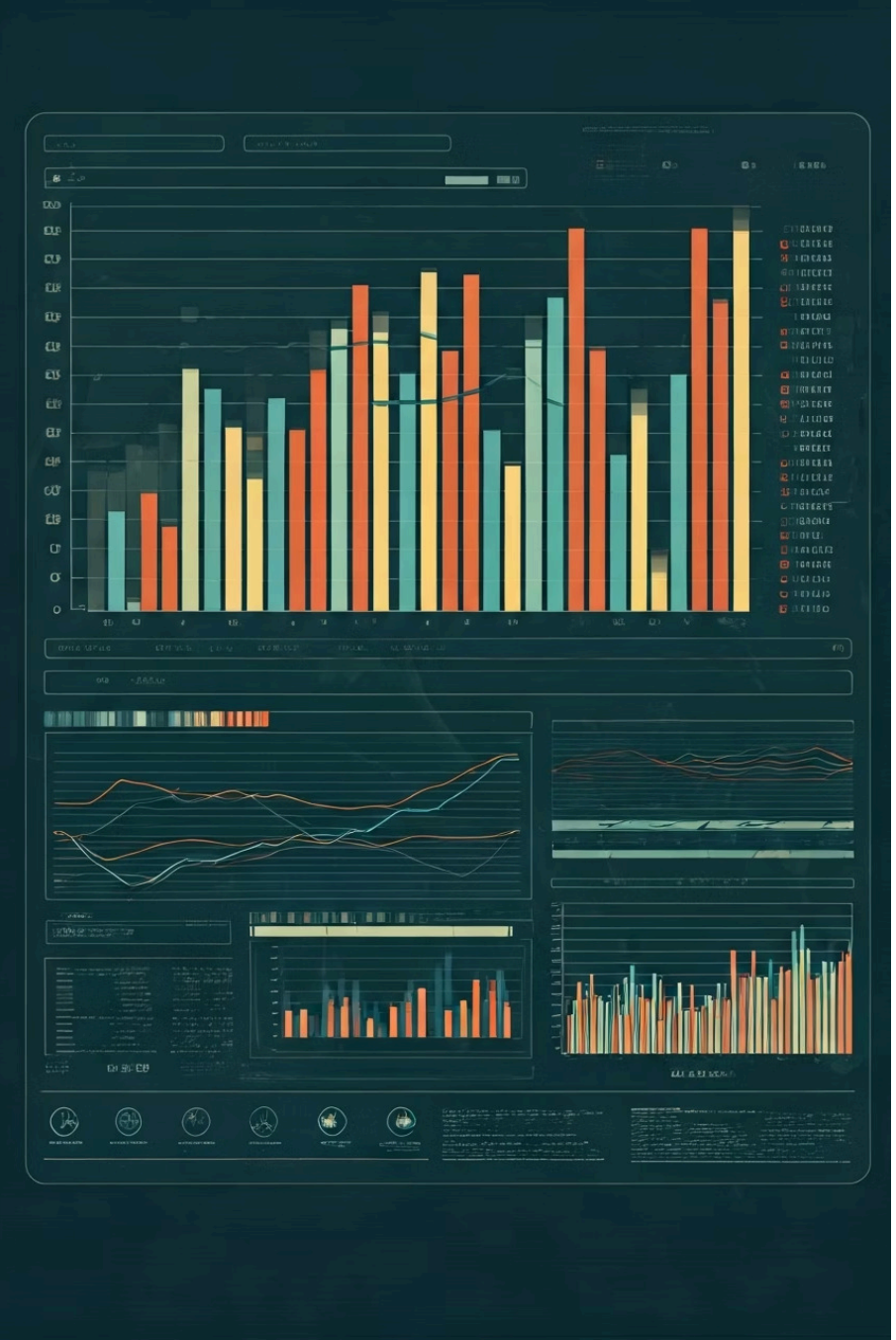

Data Visualization

Sales Trend

Line plots visualize sales progression over time.

Product Comparison

Bar charts compare performance across product categories.

Correlation Analysis

Heatmaps reveal relationships between variables.

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Key Insights

Peak Sales Months

Certain months consistently show higher sales volumes.

Product Dominance

A few product categories significantly contribute to overall revenue.

Customer Impact

Customer behavior directly influences sales performance.

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Recommendations & Conclusion

Recommendations

1. Increase stock during peak months.
2. Promote high-performing products.
3. Offer targeted discounts.
4. Focus on customer retention strategies.

Conclusion

- Successfully performed EDA on retail data.
- Generated actionable business insights.
- Improved understanding of sales patterns.

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